Lec 13. Bayesian Inference

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Bayes

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Learning objectives

After this section, you should be able to ...

- understand the basic of Bayesian inference
- conduct inference for specific problems

13.1 Introduction

- So far we have been using frequentist (or classical) methods. In the frequentist approach, probability is interpreted as long run frequencies. The goal of frequentist inference is to create procedures with long run guarantees. Indeed, a better name for frequentist inference might be procedural inference. Moreover, the guarantees should be uniform over θ if possible. For example, a confidence interval traps the true value of θ with probability $1 - \alpha$, no matter what the true value of θ is.

- In frequentist inference, procedures are random while parameters are fixed, unknown quantities

- In the Bayesian approach, probability is regarded as a measure of subjective degree of belief. In this framework, **everything, including parameters, is regarded as random**. There are no long run frequency guarantees.

- The Bayesian uses formal probability models to express personal uncertainty about that fixed unknown parameter's value. The **randomness** in these models represents personal uncertainty about the parameter's value; it is not a property of the parameter(although we should hope it accurately reflects properties of the mechanisms that produced the parameter).

- One very important point, which causes a lot of confusion, is this:

Using Bayes' Theorem $\neq$ Bayesian inference

- The difference between Bayesian inference and frequentist inference is the **goal**.
  - Bayesian Goal: Quantify and analyze subjective degrees of belief.
  - Frequentist Goal: Create procedures that have frequency guarantees.
- Neither method of inference is right or wrong. Which one you use depends on your goal. If your goal is to quantify and analyze your subjective degrees of belief, you should use Bayesian inference. If our goal create procedures that have frequency guarantees then you should use frequentist procedures.

- Sometimes you can do both. That is, sometimes a Bayesian method will also have good frequentist properties. Sometimes it won't.

- A summary of the main ideas is in Table 1.

Table 1: Bayesian versus Frequentist Inference

	Bayesian	Frequentist
Probability	subjective degree of belief	limiting frequency
Goal	analyze beliefs	create procedures with frequency guarantees
θ	random variable	fixed
X	random variable	random variable
Use Bayes' theorem?	Yes. To update beliefs.	Yes, if it leads to procedure with good frequentist behavior. Otherwise no.

13.2 Characteristics of Bayesian inference

Recap

- Firstly, Bayesian inference depends directly on the subjective definition of probability. We can all have our own probabilities for a given event A, $P(A)$. This is called **prior**.

- Secondly, given another event, say B, occurred, we update our belief of A by using Bayes' Theorem:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B|A)P(A) + P(B|\bar{A})P(\bar{A})}$$

- Similarly, we have a prior distribution $\pi(\boldsymbol{\theta})$ for the parameter $\boldsymbol{\theta}$
- Then given a sample $\mathbf{x}$, we can update our beliefs using Bayes theorem:

$$\begin{aligned}\pi(\boldsymbol{\theta} \mid \mathbf{x}) &= \frac{f(\mathbf{x} \mid \boldsymbol{\theta})\pi(\boldsymbol{\theta})}{f(\mathbf{x})} \\ &\propto f(\mathbf{x} \mid \boldsymbol{\theta})\pi(\boldsymbol{\theta}) = L(\boldsymbol{\theta} \mid \mathbf{x})\pi(\boldsymbol{\theta})\end{aligned}$$

where $f(\mathbf{x}) = \int_{\Theta} f(\mathbf{x} \mid \boldsymbol{\theta})\pi(\boldsymbol{\theta})d\boldsymbol{\theta}$. i.e.,

$$\textbf{Posterior} \propto \textbf{Likelihood} \times \textbf{Prior}$$

- Bayesian inference for a parameter is based on its posterior distribution.
- Problems:
  - Specify the prior $\pi(\boldsymbol{\theta})$
  - Computing the marginal distribution $f(\mathbf{x} \mid \boldsymbol{\theta})$

13.3 The Prior distribution

- **Subjective prior**
- **Objective prior**
  - we don't know anything about the problem at hand,
  - we would like to be *objective*.

In such situations, we should choose a non-informative prior distribution. However, there are many possible selections. Which is the most useful?

Objective prior

- **Uniform priors**: $\pi(\theta) \propto 1/|\Theta|$. This is fine in finite dimensional problems, but if the parameter space Θ is unbounded, then such prior distributions are improper($\int_\Theta \pi(\theta)d\theta = \infty$).

- **Improper prior**: $\pi(\theta) \geq 0, \int_\Theta \pi(\theta)d\theta = \infty$ such that $\pi(\theta|\mathbf{x})$ is proper.

  - $\pi(\theta) \equiv 1$ for location parameter
  - $\pi(\sigma) \propto 1/\sigma$ for scale parameter

- **Jeffreys Prior**: its density function is proportional to the square root of the determinant of the Fisher information matrix,

$$\pi(\boldsymbol{\theta}) \propto \sqrt{\det I(\boldsymbol{\theta})}$$

  which is invariant under reparametrization.

- Other non-informative priors

13.4 Bayesian computing

- One problem in the implementation of Bayesian approaches is analytical tractability. For a likelihood function $L(\boldsymbol{\theta} \mid \mathbf{x})$ and prior distribution $\pi(\boldsymbol{\theta})$, in order to calculate the posterior distribution, it is necessary to evaluate

$$f(\mathbf{x}) = \int L(\boldsymbol{\theta} \mid \mathbf{x})\pi(\boldsymbol{\theta})d\boldsymbol{\theta}$$

  and to make predictions, we must evaluate

$$f(z \mid \mathbf{x}) = \int f(z \mid \boldsymbol{\theta})\pi(\boldsymbol{\theta} \mid \mathbf{x})d\boldsymbol{\theta}$$

- **Solutions**:
  - such integrals may be evaluated numerically: MCMC
  - such integrals may be evaluated analytically: conjugate priors

Conjugate prior

Definition

If $\mathcal{F} = \{f(x|\boldsymbol{\theta}), \boldsymbol{\theta} \in \Theta\}$ is a class of sampling distributions and $\Pi = \{\pi(\boldsymbol{\theta})\}$ is a class of prior distributions for $\boldsymbol{\theta}$, we say that Π is conjugate to $\mathcal{F}$, if $\pi(\boldsymbol{\theta}|x) \in \Pi$ for all $f(\cdot|\boldsymbol{\theta}) \in \mathcal{F}$ and $\pi(\cdot) \in \Pi$.

- Binomial-Beta, Negative Binomial-Beta, Geometric-Beta
- Poisson-Gamma
- Multinomial-Direchlet
- Normal-Normal for the mean, Normal-Inverse Gamma for the variance
- Multivariate normal-normal-Wishart
- Exponential-Gamma

13.5 Bayesian Estimation

Point Estimation

- For a Bayesian, estimation is treated as a decision problem. In a given situation, we should select an estimator in order to minimize the loss that we expect to incur (discussed later).

- Based on the posterior distribution, a point estimator for θ can be

$$\hat{\theta}_B(\mathbf{x}) = \begin{cases} \hat{\theta}_{MAP} = \arg\max_\theta \pi(\theta|\mathbf{x}), & \text{Maximum A Posteriori} \\ \hat{\theta}_E = E[\theta|\mathbf{x}], & \text{Posterior Expectation} \\ \hat{\theta}_{ME} = \arg\max_a E[|\theta - a||\mathbf{x}], & \text{Posterior Median,}\theta \text{ is scalar} \end{cases}$$

  $\hat{\theta}_E$ is usually more reasonable than $\hat{\theta}_{MAP}$ (see example 7.3.2)

- The Posterior mean square error for $\hat{\theta}_B(\mathbf{x})$ is defined by

$$PMSE(\hat{\theta}_B(\mathbf{x})) = E_{\theta|\mathbf{x}}[(\theta - \hat{\theta}_B(\mathbf{x}))^2]$$

- When $\hat{\theta}_B(\mathbf{x}) = E(\theta|\mathbf{x})$, then the PMSE of $\hat{\theta}_B(\mathbf{x})$ is

$$PMSE(\hat{\theta}_B(\mathbf{x})) = E_{\theta|\mathbf{x}}[(\theta - \hat{\theta}_B(\mathbf{x}))^2] = Var(\theta|\mathbf{x}).$$

- Denote by $E(\theta|x) = \mu^\pi(x)$ the posterior mean for θ, and $V^\pi(x) = Var(\theta|x)$ the posterior variance, then

$$\begin{aligned} PMSE(\delta(x)) &= E_{\theta|x}[(\theta - \mu^\pi(x)) + (\mu^\pi(x) - \delta(x))]^2 \\ &= V^\pi(x) + (\mu^\pi(x) - \delta(x))^2 \geq V^\pi(x). \end{aligned}$$

  where the equality holds iff $\delta(x) = \mu^\pi(x)$. i.e., the posterior mean estimator $\hat{\theta}_E$ for θ minimizes the PMSE.

- $\hat{\theta}_E$ is the optimal estimator under PMSE, and thus is recommended to use in application.

Interval Estimation

Definition

A Bayesian credible interval of size $1-\alpha$ is an interval (a, b) such that

$$P(a \leq \theta \leq b|\mathbf{x}) = 1 - \alpha.$$

$$i.e., \quad \int_a^b \pi(\theta|\mathbf{x})d\theta = 1 - \alpha.$$

- Conceptually, probability comes into play in a frequentist confidence interval **before** collecting the data. Ex: there is a 95% probability that we will collect data that produces an interval that contains the true parameter value. Awkard!

- Meanwhile, probability comes into play in a Bayesian credible interval **after** collecting the data. Ex: based on the data, we now think there is a 95probability that the true parameter value is in the interval.

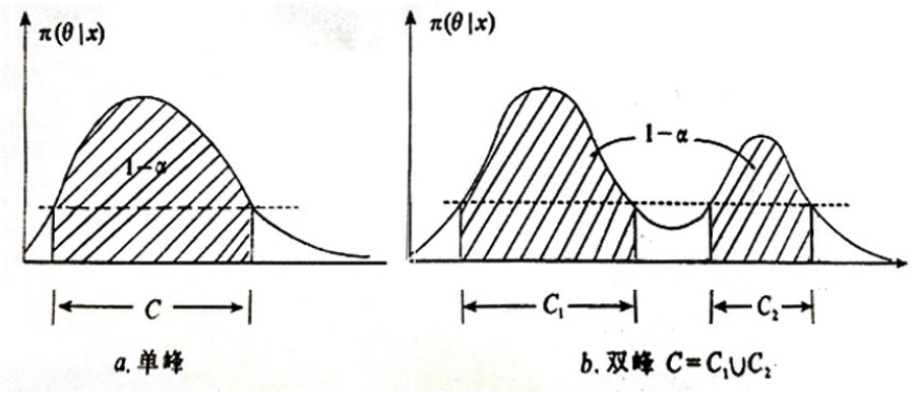
- It is clear that in general, we will have (infinitely) many credible intervals for θ. The shortest possible credible interval is called a **highest posterior density (HPD) interval**.

Definition

The $100 \times (1-\alpha)\%$ HPD interval is an interval of form

$$C = \{\theta : \pi(\theta|\mathbf{x}) \geq c(\alpha)\}$$

where $c(\alpha)$ is the largest number such that $P(\theta \in C|\mathbf{x}) \geq 1-\alpha$.



13.6 Bayesian testing

- Assume now that we wish to test the hypothesis

$$H_0 : \boldsymbol{\theta} \in \Theta_0 \leftrightarrow H_1 : \boldsymbol{\theta} \in \Theta_1$$

  where $\Theta_0 \cup \Theta_1 = \Theta$ and $\Theta_0 \cap \Theta_1 = \phi$. In theory, this is straightforward.

- Given a sample of data, $\mathbf{x}$, we can calculate the posterior probabilities $\alpha_0 = P(H_0 \mid \mathbf{x})$ and $\alpha_1 = P(H_1 \mid \mathbf{x})$, and we can reject the null hypothesis H_0 if $\alpha_1 > \alpha_0$.

- Or we can use the **Bayes factor**(the support of H_0):

$$BF_{01}(\mathbf{x}) = \frac{Posterior\ Odds}{Prior\ Odds} = \frac{\alpha_0/\alpha_1}{\pi_0/\pi_1} = \frac{\alpha_0\pi_1}{\alpha_1\pi_0}$$

  where $\pi_0 = P(H_0)$ and $\pi_1 = P(H_1)$. It tells us about the changes in our relative beliefs about the two models caused by the data. Jefferys recommends to attain H_0 when $BF_{01}(\mathbf{x}) > 3$

13.7 Bayesian Prediction

- Let $X \sim f(x|\theta)$, given a sample $\mathbf{x}$, we want to predict the value of a random variable Z which has density $g(z|\theta)$. It is commonly assumed that Z and X are independent given θ.

- Bayesian prediction logic: Let $\pi(\theta|\mathbf{x})$ be the posterior density for θ, then $g(z|\theta)\pi(\theta|\mathbf{x})$ is the joint density for (Z, θ) given $X = \mathbf{x}$, thus $p(z|\mathbf{x})$ can be obtained by integrating out θ.

Definition

Suppose X has density $f(x|\theta)$, and θ has prior $\pi(\theta)$. Given a sample $X = \mathbf{x}$, the posterior predictive density of a random variable Z is defined as

$$p(z|\mathbf{x}) = \int_{\Theta} g(z|\theta)\pi(\theta|\mathbf{x})d\theta.$$

- Point prediction: use the mean, median or mode of $p(z|\mathbf{x})$ to estimate z

- Interval prediction: use the $1-\alpha$ credible interval $[a, b]$ based on $p(z|\mathbf{x})$:

$$P(a \leq Z \leq b|\mathbf{x}) = \int_a^b p(z|\mathbf{x})dz = 1 - \alpha$$